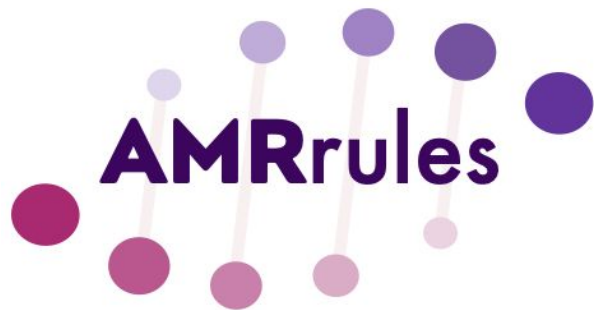


# ESGEM-AMR



ESCMID



[amrrules.org](https://amrrules.org)

# Agenda

## 1. Updates to AMRrules interpretation engine

## 2. Updates to rule specification

## 3. Evidence grades

Updates to definition

Quantitative criteria

## 4. Manuscripts & Planning

## 5. Other business

## Engine v1.1

1. Copy number rule implementation
2. Combination rule reporting changes
3. Reporting unknown/none rules (summary output)
4. Other minor updates

| Name            | Subclass     | Method | variation type                                 | gene | mutation       | ruleID  | drug         | phenotype   | clinical category | evidence grade | organism                 |
|-----------------|--------------|--------|------------------------------------------------|------|----------------|---------|--------------|-------------|-------------------|----------------|--------------------------|
| GCA_001661085.1 | AZITHROMYCIN | POINTN | Nucleotide variant detected in multi-copy gene | 23S  | c.2059A>G      | NGO0005 | azithromycin | nonwildtype | S                 | moderate       | s__Neisseria gonorrhoeae |
| GCA_001661085.1 | AZITHROMYCIN | POINTN | Nucleotide variant detected in multi-copy gene | 23S  | c.2059A>G      | NGO0005 | azithromycin | nonwildtype | S                 | moderate       | s__Neisseria gonorrhoeae |
| GCA_001661085.1 | AZITHROMYCIN | POINTN | Nucleotide variant detected in multi-copy gene | 23S  | c.2059A>G      | NGO0005 | azithromycin | nonwildtype | S                 | moderate       | s__Neisseria gonorrhoeae |
| GCA_001661085.1 | AZITHROMYCIN | POINTN | Nucleotide variant detected in multi-copy gene | 23S  | c.2059A>G      | NGO0005 | azithromycin | nonwildtype | S                 | moderate       | s__Neisseria gonorrhoeae |
| GCA_001661085.1 | -            | -      | Nucleotide variant detected in multi-copy gene | 23S  | c.[2059A>G][4] | NGO0002 | azithromycin | nonwildtype | R                 | high           | s__Neisseria gonorrhoeae |

If setting copy number rules, ensure there is a rule for a single copy for this drug/class  
e.g. c.2059A>G OR c.[2059A>G][1]

| drug          | drug class                 | clinical category | phenotype   | evidence grade | markers (non-S)    | markers (no rule) | markers (S)                         | supporting rules    | other rules         |
|---------------|----------------------------|-------------------|-------------|----------------|--------------------|-------------------|-------------------------------------|---------------------|---------------------|
| ciprofloxacin | fluoroquinolone antibiotic | R                 | nonwildtype | high           | gyrA:p.Ser91Phe    | -                 | gyrA:p.Asp95Gly;<br>parC:p.Ser87Arg | NGO0018             | NGO0022;<br>NGO0057 |
| azithromycin  | macrolide antibiotic       | R                 | nonwildtype | high           | 23S:c.[2059A>G][4] | -                 | mtrR:c.-53Adel                      | NGO0002             | NGO0005;<br>NGO0036 |
| rifampin      | rifamycin antibiotic       | -                 | -           | none           | -                  | rpoB:p.His553Asn  | -                                   | none (default none) | -                   |
| (all)         | sulfonamide antibiotic     | R                 | nonwildtype | moderate       | folP:p.Arg228Ser   | -                 | -                                   | NGO0014             | -                   |

Final marker matches final genotype call

Supporting ruleID is for the multi-copy rule (NGO0002)

If **--list-all-rules** provided, all other rules encountered are given in the 'other rules' column

| Subclass    | Method | variation type           | gene                                    | mutation    | ruleID  | drug             | phenotype   | clinical category | evidence grade | organism                 |
|-------------|--------|--------------------------|-----------------------------------------|-------------|---------|------------------|-------------|-------------------|----------------|--------------------------|
| BETA-LACTAM | POINTX | Protein variant detected | porB1b                                  | p.Ala121Asn | NGO0098 | benzylpenicillin | nonwildtype | I                 | low            | s__Neisseria gonorrhoeae |
| BETA-LACTAM | POINTX | Protein variant detected | porB1b                                  | p.Gly120Lys | NGO0094 | benzylpenicillin | nonwildtype | I                 | high           | s__Neisseria gonorrhoeae |
| -           | -      | Combination              | porB1b:p.Gly120Lys & porB1b:p.Ala121Asn | -           | NGO0123 | benzylpenicillin | nonwildtype | I                 | high           | s__Neisseria gonorrhoeae |

Addition of row for copy number rule, set all AMRFP columns to ‘-’

**Gene column becomes the rule logic, for the markers that have been detected**

| drug             | drug class                          | clinical category | phenotype   | evidence grade | markers (non-S)                                                                                                               | markers (no rule) | markers (S)                                                | supporting rules                                               | other rules                                                                                                           |
|------------------|-------------------------------------|-------------------|-------------|----------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| benzylpenicillin | beta-lactamase sensitive penicillin | I                 | nonwildtype | high           | ponA:p.Leu421Pro;<br>penA:p.Gly545Ser;<br>penA:p.Ile312Met;<br>penA:p.Val316Thr;<br>porB1b:p.Ala121Asn;<br>porB1b:p.Gly120Lys | -                 | mtrR:c.-53Adel                                             | NGO0074;<br>NGO0075;<br>NGO0089;<br>NGO0093;<br><b>NGO0123</b> | NGO0042; <b>NGO0094</b> ;<br><b>NGO0098</b>                                                                           |
| (all)            | cephalosporin                       | R                 | nonwildtype | high           | penA:p.Ala311Val;<br>penA:p.Gly545Ser;<br>penA:p.Ile312Met;<br>penA:p.Thr483Ser;<br>penA:p.Val316Thr                          | -                 | penA:p.Ala510Val;<br>penA:p.Phe504Leu;<br>penA:p.Asn512Tyr | <b>NGO0120</b> ;<br><b>NGO0121</b>                             | <b>NGO0065</b> ; <b>NGO0066</b> ;<br><b>NGO0067</b> ; <b>NGO0070</b> ;<br>NGO0082;NGO0083;<br>NGO0084; <b>NGO0087</b> |

All markers listed individually in relevant columns

**Combination rules** being applied listed in 'supporting rules'

**Components** of those combination rules listed in 'other rules'

| ruleID  | organism                               | gene ↓  | nucleotide acce... | mutation ⓘ | variation type ⓘ | gene context | drug                         | drug class                             | phenotype   | clinical category | evidence grade... |
|---------|----------------------------------------|---------|--------------------|------------|------------------|--------------|------------------------------|----------------------------------------|-------------|-------------------|-------------------|
| EFC0050 | <a href="#">Enterococcus faecalis</a>  | unknown | -                  | -          | -                | core         | <a href="#">fusidic acid</a> | -                                      | wildtype    | R                 | high              |
| EFC0051 | <a href="#">Enterococcus faecalis</a>  | unknown | -                  | -          | -                | core         | -                            | <a href="#">sulfonamide antibiotic</a> | wildtype    | R                 | high              |
| EFM0027 | <a href="#">Enterococcus faecium</a>   | unknown | -                  | -          | -                | core         | <a href="#">fusidic acid</a> | -                                      | wildtype    | R                 | high              |
| EFM0028 | <a href="#">Enterococcus faecium</a>   | unknown | -                  | -          | -                | core         | -                            | <a href="#">sulfonamide antibiotic</a> | wildtype    | R                 | high              |
| NGO0124 | <a href="#">Neisseria gonorrhoeae</a>  | unknown | -                  | -          | -                | core         | <a href="#">trimethoprim</a> | -                                      | nonwildtype | R                 | -                 |
| NMN0001 | <a href="#">Neisseria meningitidis</a> | unknown | -                  | -          | -                | core         | <a href="#">trimethoprim</a> | -                                      | wildtype    | R                 | high              |
| PRM0003 | <a href="#">Proteus mirabilis</a>      | unknown | -                  | -          | -                | core         | <a href="#">polymyxin B</a>  | -                                      | wildtype    | R                 | high              |
| PRM0004 | <a href="#">Proteus mirabilis</a>      | unknown | -                  | -          | -                | core         | <a href="#">colistin</a>     | -                                      | wildtype    | R                 | high              |
| PRM0006 | <a href="#">Proteus mirabilis</a>      | unknown | -                  | -          | -                | core         | <a href="#">imipenem</a>     | -                                      | wildtype    | I                 | high              |
| STA0001 | <a href="#">Staphylococcus aureus</a>  | unknown | -                  | -          | -                | core         | <a href="#">ceftazidime</a>  | -                                      | wildtype    | R                 | high              |

| ruleID  | organism                          | gene | nucleotide acce... | mutation ⓘ | variation type ⓘ | gene context | drug                           | drug class | phenotype | clinical category | evidence grade... |
|---------|-----------------------------------|------|--------------------|------------|------------------|--------------|--------------------------------|------------|-----------|-------------------|-------------------|
| PRM0005 | <a href="#">Proteus mirabilis</a> | none | -                  | -          | -                | -            | <a href="#">nitrofurantoin</a> | -          | wildtype  | R                 | high              |



# Unknown rules: Summary output

## *P. mirabilis* example

| drug           | drug class                   | clinical category | pheno    | evidence grade | markers (non-S)     | markers (no rule) | markers (S) | supporting rules    |
|----------------|------------------------------|-------------------|----------|----------------|---------------------|-------------------|-------------|---------------------|
| tobramycin     | aminoglycoside antibiotic    | -                 | -        | none           | -                   | ant(2'')-Ia       | -           | none (default none) |
| polymyxin B    | antibiotic mixture           | R                 | wildtype | high           | (unknown mechanism) | -                 | -           | PRM0003             |
| imipenem       | carbapenem                   | I                 | wildtype | high           | (unknown mechanism) | -                 | -           | PRM0006             |
| trimethoprim   | diaminopyrimidine antibiotic | -                 | -        | none           | -                   | dfrA1             | -           | none (default none) |
| nitrofurantoin | nitrofuran antibiotic        | R                 | wildtype | high           | (no target)         | -                 | -           | PRM0005             |

## Other minor updates

- Addition of `--list-all-rules` argument
  - ruleIDs now given in 'supporting rules' and 'other rules' columns
  - if no rules, explicitly states 'none' and no rule interpretation default
- Interpreted output now sorted by marker
  - Allows easier identification of multi-copy or combination rules
  - Was previously in the same order as original input file
- Engine version now reported in summary output
- Quality of life improvements for testing purposes

**Engine v1.1 will be released for paper**

**We will let you know when there is a beta-release to test**

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**1. Updates to AMRRules interpretation engine**

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# New CARD release in August

Includes new ontology for 'variation type' - <https://card.mcmaster.ca/ontology/48425>

## ARO variant type

[Download Sequences](#)

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ontology       | CARD's Antibiotic Resistance Ontology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Accession      | ARO:3009567                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Definition     | Type of variation reported by resistance gene annotators or interpreters. Based on the schemes developed by hAMRization ( <a href="https://github.com/pha4ge/hAMRization">https://github.com/pha4ge/hAMRization</a> ) and AMRRules ( <a href="https://docs.amrrules.org">https://docs.amrrules.org</a> ), with additional terms from the NCIT ontology.                                                                                                                                                                                                                                                                                                             |
| Classification | <b>1 ontology terms</b>   <a href="#">Show</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Parent Term(s) | <b>1 ontology terms</b>   <a href="#">Show</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Sub-Term(s)    | <b>10 ontology terms</b>   <a href="#">Hide</a> <ul style="list-style-type: none"><li>+ <a href="#">gene presence detected</a></li><li>+ <a href="#">protein variant detected</a></li><li>+ <a href="#">nucleotide variant detected</a></li><li>+ <a href="#">promoter variant detected</a></li><li>+ <a href="#">disrupting variant detected</a></li><li>+ <a href="#">gene truncation detected</a></li><li>+ <a href="#">gene copy number variant detected</a></li><li>+ <a href="#">nucleotide variant detected in multi-copy gene</a></li><li>+ <a href="#">low frequency variant detected</a></li><li>+ <a href="#">combination of variant types</a></li></ul> |

# New CARD release in August

Includes new ontology for 'variation type' - <https://card.mcmaster.ca/ontology/48425>

## disrupting variant detected

[Download Sequences](#)

|            |                                                                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ontology   | CARD's Antibiotic Resistance Ontology                                                                                                                                                         |
| Accession  | NCIT:C178119                                                                                                                                                                                  |
| Synonym(s) | <i>inactivating mutation detected</i>                                                                                                                                                         |
| Definition | The gene (or protein) has been detected in the query but includes a loss of function mutation (e.g., frameshift, internal stop, deletion, truncation).<br>Paraphrased from the NCI Thesaurus. |

AMRrules v1 spec used the term:  
***'inactivating mutation detected'***

In CARD ontology, rephrased to:  
***'disrupting variant detected'***

We will update the fields in  
AMRrules v1.1 and technical  
guidance.

- Note that 'inactivating mutation detected' remains a valid synonym in CARD
- Also note both terms map to NCIT:C178119 'Loss of function gene mutation' with synonyms 'Loss of function mutation', 'Inactivating gene mutation', 'Inactivating mutation'.

## ARO, mechanism, and gene family

- Discussed in previous meetings
- ARO was already entered by subgroups for many v1 rules
  - We have now filled the gaps
  - Some variants lack ARO terms, these are left blank
- ARO used to import 'mechanism' and 'gene family' for v1 rules, where possible
- Where no ARO, 'mechanism' and 'gene family' need to be added manually by subgroups please

***Subgroups to please review these updates, and fill gaps, for inclusion in v1.1 for paper***

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## Evidence grades

- Discussed in July meeting, agreed to take this forward to v1.1
- This is now updated in ESGEM-AMR guidance, AMRrules template & docs

*Quick recap*



## Evidence grades: what they mean

|                                                                                | high           | moderate             | low                       | very low               | insufficient evidence                      |
|--------------------------------------------------------------------------------|----------------|----------------------|---------------------------|------------------------|--------------------------------------------|
| <b>How confident are you in the interpretation?</b>                            | very confident | moderately confident | it is most likely correct | very little confidence | not at all, cannot offer an interpretation |
| <b>Will the interpretation change with future research?</b>                    | very unlikely  | unlikely             | maybe                     | likely                 | yes                                        |
| <b>Will your confidence in the interpretation change with future research?</b> | very unlikely  | likely               | yes                       | yes                    | yes                                        |

## Evidence grades: specific criteria

|                                                                                                                                                                               | high           | moderate                              | low                                                            | low                                                    | very low                                                           | very low                                                     | insufficient evidence                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Types of evidence:</b> <ul style="list-style-type: none"> <li>• Experimental evidence</li> <li>• Statistical genotype/phenotype evidence in natural populations</li> </ul> | Both available | Both available                        | Only one available                                             | Both available                                         | Only one available                                                 | Both available                                               | One or both unavailable                                                                       |
| <b>Key dimensions:</b> <ul style="list-style-type: none"> <li>• precision</li> <li>• consistency</li> <li>• directness</li> <li>• risk of bias</li> </ul>                     | No concerns    | Rated down on ONE of these dimensions | For the evidence available, no concerns about these dimensions | Both types of evidence rated down on ONE key dimension | For the evidence available, concerns about MULTIPLE key dimensions | Both types of evidence rated down on MULTIPLE key dimensions | None available, or Available evidence insufficient to support one interpretation over another |
| <b>Will the interpretation change with future research?</b>                                                                                                                   | very unlikely  | unlikely                              | maybe                                                          | maybe                                                  | likely                                                             | likely                                                       | yes                                                                                           |

## Evidence grades

- Discussed in July meeting, agreed to take this forward to v1.1
- This is now updated in ESGEM-AMR guidance, AMRrules template & docs
- **Subgroups to please review their current rules and make any required updates for inclusion in v1.1**
  - e.g. bump down “very low” grades to “insufficient evidence”
  - Note that core genes will usually have good epidemiological evidence because they are intrinsic and high penetrance, so impact will be more significant on acquired genes

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# How should quantitative thresholds map to evidence grades and evidence limitations?

Quantitative values from statistical geno/pheno analysis in natural populations depend on:

- Quality of phenotype data
- Quality of genotype data
- Sample size
- Sample diversity

As per evidence grade definitions, need to consider key dimensions:

- Precision (sample size and data quality impact power and precision)
- Consistency (evidence from different studies, assay methods)
- Directness (genomic confirmation of species, per-drug analysis)
- Risk of bias (genetic/geographic/temporal diversity of strains, number of independent contributing studies/labs, calibration of automated platforms)

# How should quantitative thresholds map to evidence grades and evidence limitations?

**Can we be guided by established criteria for AST assessment?**

US FDA:

- Categorical agreement >90%
- Major error (ME) <3% (test calls susceptible isolate as 'R')
- Very major error (VME) <1.5% (test calls resistant isolate as 'S')

ISO 20776-2 (for qualitative AST devices)

- Sensitivity and specificity  $\geq 95\%$

Note that we are grading evidence for specific markers/combinations, not an overall interpretation from of genotype (which is the product of applying multiple rules)... so it is not possible or relevant to calculate these metrics directly for an individual rule

**But, they provide a baseline for the expected errors in the AST phenotype data used in the calculation**

# How to: the application and analysis of categorical agreement in antimicrobial susceptibility testing using European Committee on Antimicrobial Susceptibility Testing breakpoints

John Turnidge <sup>1,2</sup> · Gunnar Kahlmeter <sup>2,3</sup> · Christian G. Giske <sup>4,5</sup> · Rafael Cantón <sup>6,7</sup> · Erika Matuschek <sup>3</sup> · Mandy Wootton <sup>8</sup> · Sören Gatermann <sup>9</sup> Show less

|                          |                                          | Reference/Standardised or 'True' category |                                     |               |
|--------------------------|------------------------------------------|-------------------------------------------|-------------------------------------|---------------|
|                          |                                          | Susceptible, Standard dosing regimen (S)  | Susceptible, Increased Exposure (I) | Resistant (R) |
| Test and report category | Susceptible, Standard dosing regimen (S) |                                           | (MD)                                | VMD           |
|                          | Susceptible, Increased exposure (I)      | (mD)                                      |                                     | VMD           |
|                          | Resistant (R)                            | (MD)                                      | (MD)                                |               |

**Table 1**  
New EUCAST interpretations of categorical agreement

| 'True' susceptibility category | Tested and reported susceptibility category | Characterization of discrepancy | Rationale                                                                             | Potential consequences                                                                            |
|--------------------------------|---------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| S                              | S                                           | No discrepancy                  | Correct report                                                                        | None                                                                                              |
| I                              | I                                           | No discrepancy                  | Correct report                                                                        | None                                                                                              |
| R                              | R                                           | No discrepancy                  | Correct report                                                                        | None                                                                                              |
| S                              | I                                           | mD                              | Both are susceptible categories; the consequence is an unnecessary increased exposure | Increased risk of adverse reactions; Increased cost of agent and/or administration                |
| S                              | R                                           | MD                              | False resistance removes a therapeutic option                                         | Therapeutic alternative(s) may be more expensive and/or more toxic                                |
| I                              | S                                           | MD                              | Both are susceptible categories; the consequence is suboptimal antimicrobial exposure | Increased risk of therapeutic failure and resistance selection                                    |
| I                              | R                                           | MD                              | False resistance removes a therapeutic option                                         | Therapeutic alternative(s) may be more expensive and/or more toxic                                |
| R                              | S                                           | VMD                             | False susceptibility may have serious consequences                                    | Risk of continuing or switching to ineffective therapy with increased risk of therapeutic failure |
| R                              | I                                           | VMD                             | False susceptibility may have serious consequences                                    | Risk of continuing or switching to ineffective therapy with increased risk of therapeutic failure |

EUCAST, European Committee on Antimicrobial Susceptibility Testing; I, susceptible, increased exposure; MD, major discrepancy; mD, minor discrepancy; R, resistant; S, susceptible, standard dosing regimen; VMD, very major discrepancy.

New EUCAST paper shared by Rafael Cantón

(we will add these new discrepancy definitions to the proposed metrics in the slides that follow - but will need to discuss appropriate thresholds)

# To support an 'R' rule being graded 'high'

## Solo Positive Predictive Value (PPV) for R

= of those isolates with this marker, and no other known markers for this drug class, what fraction are phenotypically categorised as 'R' using MIC or disk diffusion, interpreted with the latest EUCAST breakpoints?

*This is our strongest signal to check whether resistance is directly attributable to this specific marker, i.e. whether the interpretation for detection of this marker should be set to 'R'.*

**Proposed threshold:  $\geq 90\%$** , based on the established categorical agreement threshold

*We are allowing for up to 10% of the isolates that we are analysing (with the marker found solo) to be incorrectly categorised as S/I, based on our expectation of  $\geq 90\%$  categorical agreement.*

## Positive Predictive Value (PPV) for R

= of all those isolates with this marker, whether or not they have additional markers, what fraction are R?

*This is an extra check to make sure that isolates with this marker can be reliably predicted as resistant. This value is only relevant if the solo PPV is already known to be high, as it could be inflated by mechanisms associated with other markers that are not linked to the presence of this marker.*

**Proposed threshold:  $\geq 90\%$** , based on the established categorical agreement threshold, as above



# To support an 'R' rule being graded 'high'

## Major errors (ME) attributable to this marker

= of those isolates that are actually S, how many carry the marker (with or without others) and thus would be wrongly interpreted as 'R' on the basis of this rule?

*This is not an overall 'major error rate' for genotype-based predictions for this drug, but indicates what proportion of isolates would be classified as major errors specifically because of the 'R' rule associated with this marker. The overall ME rate may be higher than this value, and so the threshold should not exceed the overall ME threshold*

**Proposed threshold:**  $\leq 3\%$ , based on the established major error threshold

*We will tolerate up to 3% of all susceptible isolates in the dataset being incorrectly categorised as R due to this rule, based on the established threshold of  $\leq 3\%$  major errors, but note that additional major errors may occur for this drug due to other rules.*

# To support an 'R' rule being graded 'high' (?)

## Adjusted Odds Ratio (aOR) for resistance in multivariable model

= In a multiple logistic regression model for R, including all markers associated with this drug class, what is the effect size (expressed as OR) for this marker?

*This is a secondary signal as to the independent effect of this marker on phenotype. A high value is insufficient to assign an interpretation of 'R', as it may be high and significant even if the marker is not directly contributing to resistance. However if the 95% CI for the OR includes one, this signifies a lack of evidence supporting the hypothesis that the marker contributes independently to resistance.*

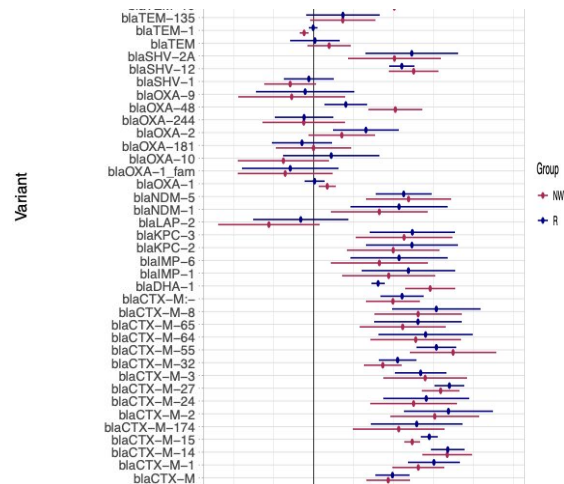
**Proposed threshold: 95% CI for aOR does not include 1 (i.e. no effect)**

## Odds Ratio (OR) for resistance in single variable model

= In a logistic regression model for R, with this marker as sole predictor, what is the effect size (expressed as OR) for this marker?

*This is a weak signal. It is unlikely to contribute additional information beyond the PPV >90%.*

**Proposed threshold: 95% CI for OR does not include one (i.e. no effect)**



# To support an 'S' rule being graded 'high'

## Solo Positive Predictive Value (PPV) for S

= of those isolates with this marker, and no other known markers for this drug class, what fraction are phenotypically categorised as 'S' using MIC or disk diffusion, interpreted with the latest EUCAST breakpoints?

*This is our strongest signal to check whether the presence of this specific marker is sufficient to predict resistance, i.e. whether the interpretation for detection of this marker should be set to 'S'.*

**Proposed threshold:  $\geq 90\%$** , based on the established categorical agreement threshold

*We are allowing for up to 10% of the isolates that we are analysing (with the marker found solo) to be incorrectly categorised as R/I, based on our expectation of  $\geq 90\%$  categorical agreement.*

# To support an 'S' rule being graded 'high'

## Very major errors (VME) attributable to this marker

= of those isolates that are actually R, how many carry this marker solo (i.e. with no other markers that are associated with resistance to this drug class) and thus would be wrongly interpreted as 'S' on the basis of this rule?

*This is not an overall 'very major error rate' for genotype-based predictions for this drug, but indicates what proportion of R isolates would be classified as very major errors that would not be VMEs if the rule for this marker was updated. The overall VME rate may be higher than this value, and so the threshold should not exceed the overall VME threshold*

**Proposed threshold:**  $\leq 1.5\%$ , based on the established major error threshold

*We will tolerate up to 1.5% of all resistant isolates in the dataset being incorrectly categorised as S due to this rule, based on the established threshold of  $\leq 1.5\%$  very major errors, but note that additional errors may occur for this drug due to other rules.*

# To support an 'S' rule being graded 'high'

(Only applies if there are <3 samples with this marker solo, and so no solo PPV)

## Adjusted Odds Ratio (aOR) for resistance in multivariable model

= In a multiple logistic regression model for I/R, including all markers associated with this drug class, what is the effect size (expressed as OR) for this marker?

*This is a secondary signal as to the independent effect of this marker on phenotype. A high value is insufficient to assign an interpretation of 'I' or 'R', as it may be high and significant even if the marker is not directly contributing to resistance. However if the 95% CI for the OR crosses zero, this is consistent with the hypothesis that the marker does not contribute independently to resistance. In the absence of solo-PPV data for this marker, the logistic regression evidence can be sufficient to grade the 'S' call as 'high' (not this would also require mechanistic evidence/reasoning)*

**Proposed threshold: 95% CI for aOR includes zero**

# To support an 'I' rule being graded 'high'

## Solo Positive Predictive Value (PPV) for I

= of those isolates with this marker, and no other known markers for this drug class, what fraction are phenotypically categorised as 'I' using MIC or disk diffusion, interpreted with the latest EUCAST breakpoints?

*This is our strongest signal to check whether the 'I' phenotype is directly attributable to this specific marker, i.e. whether the interpretation for detection of this marker should be set to 'I' rather than 'R' or 'S'.*

**Proposed threshold:  $\geq 90\%$** , based on the established categorical agreement threshold

*We are allowing for up to 10% of the isolates that we are analysing (with the marker found solo) to be incorrectly categorised as S/I, based on our expectation of  $\geq 90\%$  categorical agreement.*

*(e.g. if 5% S, 60% I, 35% R -> we interpret as 'I' but this fails to be graded 'high' because we are not sure whether to call as I or R)*

## Positive Predictive Value (PPV) for I/R

= of all those isolates with this marker, whether or not they have additional markers, what fraction are I or R?

*This is an extra check to make sure that isolates with this marker can be reliably interpreted as not S. This value is only relevant if the solo PPV is already known to be high, as it could be inflated by mechanisms associated with other markers that are not linked to the presence of this marker. Note it is not specific to PPV for R, as it is expected that the presence of additional markers will likely increase the MIC further to reach the R breakpoint, so the presence of this marker in resistant isolates does not provide evidence against the interpretation of the solo marker as 'I'.*

**Proposed threshold:  $\geq 90\%$** , based on the established categorical agreement threshold, as above

# To support an 'I' rule being graded 'high'

## Major errors (ME) attributable to this marker

= of those isolates that are actually S, how many carry the marker (with or without others) and thus would be wrongly interpreted as 'I' on the basis of this rule?

*This is not an overall 'major error rate' for genotype-based predictions for this drug, but indicates what proportion of isolates would be classified as major errors specifically because of the 'I' rule associated with this marker. The overall ME rate may be higher than this value, and so the threshold should not exceed the overall ME threshold*

**Proposed threshold:**  $\leq 3\%$ , based on the established major error threshold

*We will tolerate up to 3% of all susceptible isolates in the dataset being incorrectly categorised as I due to this rule, based on the established threshold of  $\leq 3\%$  major errors, but note that additional major errors may occur for this drug due to other rules.*

## Very major errors (VME) attributable to this marker

= of those isolates that are actually R, how many carry this marker solo (i.e. with no other markers that are associated with resistance to this drug class) and thus would be wrongly interpreted as 'I' on the basis of this rule?

*This is not an overall 'very major error rate' for genotype-based predictions for this drug, but indicates what proportion of R isolates would be classified as very major errors that would not be VMEs if the rule for this marker was updated. The overall VME rate may be higher than this value, and so the threshold should not exceed the overall VME threshold*

**Proposed threshold:**  $\leq 1.5\%$ , based on the established major error threshold

*We will tolerate up to 1.5% of all resistant isolates in the dataset being incorrectly categorised as I due to this rule, based on the established threshold of  $\leq 1.5\%$  very major errors, but note that additional errors may occur for this drug due to other rules.*

# To support an 'I' rule being graded 'high'

## **Adjusted Odds Ratio (aOR) for resistance in multivariable model**

= In a multiple logistic regression model for I/R, including all markers associated with this drug class, what is the effect size (expressed as OR) and p-value for this marker?

*This is a secondary signal as to the independent effect of this marker on phenotype. A high value is insufficient to assign an interpretation of 'I', as it may be high and significant even if the marker is not directly contributing to resistance. However if the 95% CI for the OR crosses zero, this signifies a lack of evidence supporting the hypothesis that the marker contributes independently to resistance.*

**Proposed threshold: 95% CI for aOR does not cross zero**

## **Odds Ratio (OR) for resistance in single variable model**

= In a logistic regression model for I/R, with this marker as sole predictor, what is the effect size (expressed as OR) and p-value for this marker?

*This is a weak signal. It is unlikely to contribute additional information beyond the PPV >90%.*

**Proposed threshold: 95% CI for OR does not cross zero**



# Other checks to qualify for 'high' grade

## Inconsistency

- No disagreement between MIC and disk-based calls
- No disagreement between MIC/disk data and extended (S/I/R only) dataset

## Imprecision

- No significant impact of MIC values expressed as ranges
- MIC and disk IQR do not cross the breakpoint

## Bias

- At least **3(?)** data sources contribute isolates with solo markers supporting the call
- Could also add a check for the number of STs & countries supporting the call, but suggest deficiencies on this front should be simply noted as a limitation, rather than affecting the grade
  - Fewer than 3 STs: 'lacks evidence across diverse allelic backgrounds'
  - Fewer than 3 countries: 'lacks evidence across diverse geographies'

# Checks for call of WT/NWT

**To support an 'NWT' rule being graded 'high': same criteria as for 'R'**

**To support an 'WT' rule being graded 'high': same criteria as for 'S'**

As a single grade is given for the rule, which will typically include both a clinical category (S/I/R) and phenotype (WT/NWT) interpretation, the statistical thresholds need to be met for both types of interpretation.

I.e., the overall grade assessment based on quantitative data will be the lowest grade applicable to either S/I/R or WT/NWT classification.

Note though that the S/I/R call tends to be less robust so typically an overall low grade will reflect the confidence in the category.

# Evidence grades: criteria for quantitative data

|                                                                                                  | high                            | moderate                      | low                             | very low | insufficient evidence |
|--------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|---------------------------------|----------|-----------------------|
| <b>Sample size for solo PPV</b> (i.e. number of solo-marker isolates with MIC/disk measurements) | ≥20                             | ≥10                           | ≥10                             | ≥3       | -                     |
| <b>Statistical thresholds</b> (otherwise downgraded on 'precision')                              | sPPV 90%<br>aME 3%<br>aVME 1.5% | sPPV 90%<br>aME 5%<br>aVME 5% | sPPV 80%<br>aME 10%<br>aVME 10% | sPPV 50% | -                     |
| <b>Will the interpretation change with future research?</b>                                      | very unlikely                   | unlikely                      | maybe                           | likely   | yes                   |

**Other key dimensions resulting in downgrade**  
(and note for 'evidence limitation' field)

- Inconsistency** ('conflicting evidence')
- Disagreement between MIC and disk-based calls
  - Disagreement between MIC/disk data and categorical

- Imprecision** ('lacks precision')
- Impact of MIC values expressed as ranges
  - MIC or zone IQR span the breakpoint

- Bias** ('limited sources of geno/pheno data')
- <3 data sources contributing to solo PPV

## Evidence limitations: what prevented a ‘high’ grade

|                                                                                                  | high                            | moderate                      | low                             | very low | insufficient evidence | ‘evidence limitation’ recorded if high threshold not met: |
|--------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|---------------------------------|----------|-----------------------|-----------------------------------------------------------|
| <b>Sample size for solo PPV</b> (i.e. number of solo-marker isolates with MIC/disk measurements) | ≥20                             | ≥10                           | ≥10                             | ≥3       | -                     | limited samples                                           |
| <b>Statistical thresholds</b> (otherwise downgraded on ‘precision’)                              | sPPV 90%<br>aME 3%<br>aVME 1.5% | sPPV 90%<br>aME 5%<br>aVME 5% | sPPV 80%<br>aME 10%<br>aVME 10% | sPPV 50% | -                     | lacks precision                                           |
| <b>Will the interpretation change with future research?</b>                                      | very unlikely                   | unlikely                      | maybe                           | likely   | yes                   |                                                           |

# Agenda

**1. Updates to AMRRules interpretation engine**

**2. Updates to rule specification**

**3. Evidence grades**

Updates to definition

Quantitative criteria

**4. Manuscripts & Planning**

**5. Other business**

## 2026 Manuscripts

1. **AST data sharing in genomics databases** - *accepted in npj AMR*
2. **AMRgen R package** - *submitted to Microbial Genomics*
3. **AMRrules v1** - *draft to be distributed for review this month*

### Subgroup-led manuscripts:

- *Neisseria gonorrhoeae*
- *Staphylococcus aureus*
- Grading evidence
- Others?

# Planning

## All v1 rules to be updated for inclusion in v1.1

- Review the newly added 'ARO', 'mechanism' and 'gene family' fields
  - These have been added to the new version, please check for any issues and fill in any gaps
- Review current evidence grades against the updated definitions, and make changes to grades
  - e.g. check if “very low” grades should be changed to “insufficient evidence”

## Deadline: October 31

*(for inclusion in v1.1 release to accompany the paper)*

# Planning

## Oct/Nov

- **Subgroups:**
  - Review v1.1 updates
  - Collate geno/pheno data
- **All members:** review AMRrules paper
- **Chairs:**
  - Finalise v1.1 AMRrules release (code & rules)
  - Implement quantitative evidence grades in AMRrulemakerR
  - Public geno/pheno data repository (AllTheBacteria/NCBI/EBI)

## Nov/Dec

- **Subgroups:** run AMRrulemakerR to propose rules, review & curate outputs
- **Chairs:** submit AMRrules paper

## 2027 priorities

- **AMRrules v2 release with acquired rules**
- **Validation strategies**
- **CARD compatibility**
- **WHO collaboration**

**ESCMID Global  
Meet-up?  
Hackathon?**



## Reminder

Please check the website to make sure your name is listed in the correct groups following the refresh:

<https://esgem-amr.amrrules.org/members>

Any issues email [esgem.amr@gmail.com](mailto:esgem.amr@gmail.com)

# Agenda

**1. Updates to AMRRules interpretation engine**

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# Feedback wanted: AllTheBacteria browser

<https://allthebacteria.org/browse/>



AllTheBacteria

[Home](#)

[Browse data](#)

[Resources](#)

[Docs](#)



## Browse All The Bacteria

[About the source data](#)

Try an example query: [Overview of UK samples, 2020–2024](#)

[mecA-carrying samples](#)

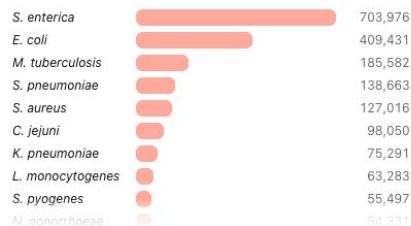
[Resistome of all \*E. coli\* samples](#)

[About the filters and charts](#)

Search by species, AMR genes, or metadata...

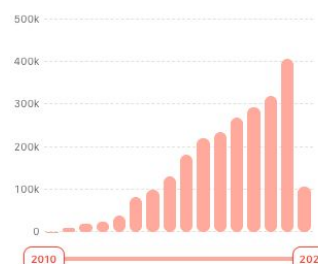
HQ samples ▾

### Species distribution

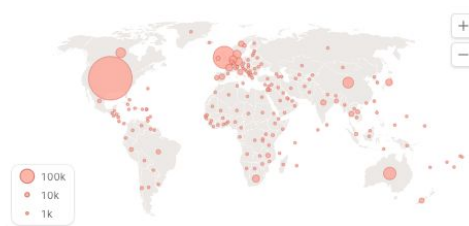


Showing top 10 species · [Show more](#)

### Year distribution



### Geographic distribution



2,465,670 samples sorted by first public

Showing 5 of 23 columns ▾ [Export](#)

| Sample         | Species                    | Country   | First public | AMR genes                                                    | Show as | List | Matrix |
|----------------|----------------------------|-----------|--------------|--------------------------------------------------------------|---------|------|--------|
| SAMEA118214836 | Mycobacterium tuberculosis | Argentina | 2025-05-05   | aac(2')-Ic blaC erm(37)                                      |         |      |        |
| SAMD00654274   | Staphylococcus aureus      | Japan     | 2025-05-04   | blaI blaR1 blaZ fosB gyrA_S84L +8                            |         |      |        |
| SAMD00654276   | Staphylococcus aureus      | Japan     | 2025-05-04   | ant(9)-Ia erm(A) gyrA_S84L mecA mecR1 +4                     |         |      |        |
| SAMD00654277   | Staphylococcus aureus      | Japan     | 2025-05-04   | ant(9)-Ia blaI blaR1 blaZ erm(A) +7                          |         |      |        |
| SAMD00654278   | Staphylococcus aureus      | Japan     | 2025-05-04   | blaI blaR1 blaZ fosB gyrA_S84L +8                            |         |      |        |
| SAMD00654296   | Staphylococcus aureus      | Japan     | 2025-05-04   | aac(6')-Ie/aph(2'')-Ia ant(6)-Ia aph(3')-IIIa blaI blaR1 +11 |         |      |        |

Contact Timo Saratto  
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# Questions? / Any other business?

**ESGEM-AMR**



**ESCMID**



[amrrules.org](https://amrrules.org)